



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2019

**HSC
Trial
Examination**

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I – 10 marks (pages 3 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7 – 17)

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1-10	11	12	13	14	15	16	Total
Mark	/10	/15	/15	/15	/15	/15	/15	/100

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What is the value of $(2+i^5)(i^2-i^3)$?
- (A) $5-5i$
- (B) $7-i$
- (C) $-1+3i$
- (D) $-3+i$
- 2 Which expression is equal to $\int x^2 \cos x \, dx$?
- (A) $x^2 \sin x + \int 2x \sin x \, dx$
- (B) $x^2 \sin x - \int 2x \sin x \, dx$
- (C) $2x \sin x - \int x^2 \sin x \, dx$
- (D) $2x \sin x + \int x^2 \sin x \, dx$
- 3 A directrix of an ellipse has the equation $x = \frac{25}{4}$ and one of its foci has the coordinates $(-4,0)$. What is the equation of the ellipse?
- (A) $\frac{x^2}{5} + \frac{y^2}{3} = 1$
- (B) $\frac{x^2}{3} + \frac{y^2}{5} = 1$
- (C) $\frac{x^2}{25} + \frac{y^2}{9} = 1$
- (D) $\frac{x^2}{9} + \frac{y^2}{25} = 1$

4 Given the substitution $x = \pi - y$, which of the following is equal to $\int_0^{\pi} x \sin x \, dx$?

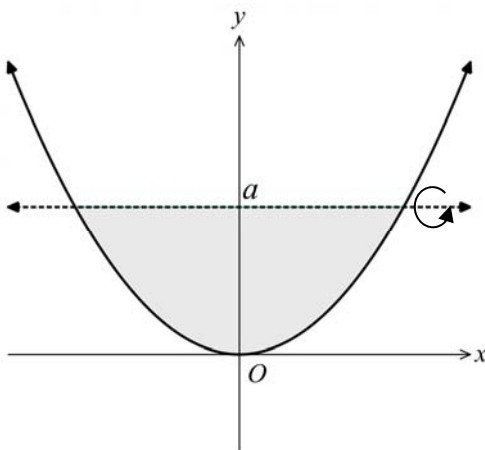
(A) $\int_{-\pi}^{\pi} \sin x \, dx$

(B) $\frac{\pi}{2} \int_0^{\pi} \sin x \, dx$

(C) $\pi \int_0^{\pi} \sin x \, dx$

(D) $\int_0^{\pi} \sin x \, dx$

5 The region bounded by the parabola $x^2 = 4ay$ and the line $y = a$ is rotated about the line $y = a$ to form a solid.



Which expression represents the volume of the solid?

(A) $2\pi \int_0^{2a} \left(a - \frac{x^2}{4a} \right)^2 dx$

(B) $2\pi \int_0^{2a} \left(a^2 - \left(\frac{x^2}{4a} \right)^2 \right) dx$

(C) $\pi \int_0^{2a} \left(a - \frac{x^2}{4a} \right)^2 dx$

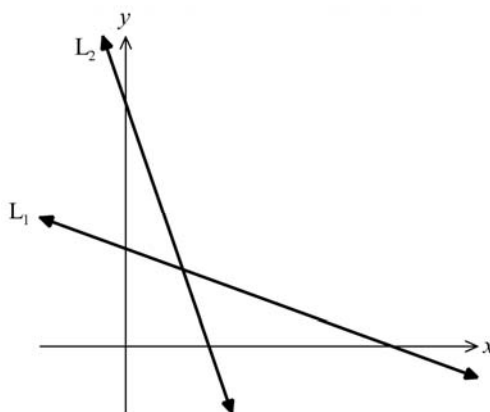
(D) $\pi \int_0^{2a} \left(a^2 - \left(\frac{x^2}{4a} \right)^2 \right) dx$

- 6 Let e be the eccentricity of a conic, centred at the origin, with both foci on the x -axis. Which of the following is NOT true?
- (A) If two ellipses have the same foci and directrices, then they have the same equation.
- (B) If two hyperbolae have equal eccentricity, then they share the same asymptotes.
- (C) For the hyperbola, as $e \rightarrow \infty$, the asymptotes approach the x -axis.
- (D) For the ellipse, as $e \rightarrow 0$, the directrices move further away from the origin whilst the foci approach the origin.
- 7 Which complex number is a root of $z^6 + i = 0$?
- (A) $-1 - i$
- (B) $-1 + i$
- (C) $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$
- (D) $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$
- 8 If the tangent at the point $(2 \sec \phi, 3 \tan \phi)$ on the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$ is parallel to $3x - y + 4 = 0$, then what is a possible value of ϕ ?
- (A) 45°
- (B) 60°
- (C) 30°
- (D) 75°

- 9 Let the complex number z satisfy the equation $|z + 4i| = 3$. What are the greatest and least values of $|z + 3|$?

- (A) 8 and 2
- (B) 5 and 2
- (C) 8 and 3
- (D) 8 and 5

- 10 The diagram below shows the graphs of the straight lines L_1 and L_2 , whose equations are $y = ax + b$ and $y = cx + d$ respectively.



Which of the following are true?

- I. $c < a$
 - II. $d > b$
 - III. $ad > bc$
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III

Section II

Total marks – 90

Attempt Questions 11–16

Allow about 2 hour 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 to 16, your responses should include relevant mathematical reasoning and/or calculations.

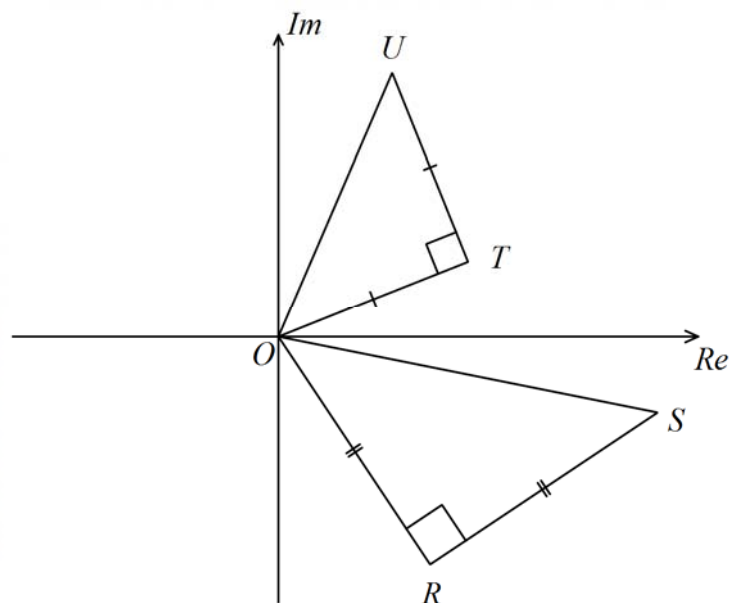
Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Express $\frac{4+3i}{2-i}$ in the form $x+iy$, where x and y are real. **2**
- (b) Consider the complex numbers $z = -1 + \sqrt{3}i$ and $w = 2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$.
- (i) Express z in modulus-argument form. **2**
- (ii) Find the argument of $\frac{z}{w}$. **1**
- (c) (i) Express $\frac{3x^2+x+6}{(x^2+9)(x-1)}$ in the form $\frac{Ax+B}{x^2+9} + \frac{C}{x-1}$. **2**
- (ii) Hence find $\int \frac{3x^2+x+6}{(x^2+9)(x-1)} dx$. **2**
- (d) The equation $|z-2| - |z+2| = \pm 2$ corresponds to a conic in the Argand diagram. **3**
Sketch the conic, showing any asymptotes, foci and directrices.
- (e) The polynomial $P(x) = x^4 + 3x^3 - x^2 - 13x - 10$ has a zero at $x = -2 - i$.
- (i) Explain why $x = -2 + i$ is also a zero. **1**
- (ii) Fully factorise $P(x)$ over the real numbers. **2**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a)



In the Argand diagram above, the points O, R, S, T and U correspond to the complex numbers $0, r, s, t$ and u respectively. The triangles ORS and OTU are right-angled isosceles triangles. Let $\omega = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$.

(i) Explain why $u = \sqrt{2} \omega t$. 1

(ii) Show that $r = \frac{s}{\sqrt{2} \omega}$. 1

(iii) Using complex numbers show that $\frac{SU}{RT} = \sqrt{2}$. 2

(b) Let $I = \int \frac{\sin x}{\sin x + 2 \cos x} dx$ and $J = \int \frac{\cos x}{\sin x + 2 \cos x} dx$.

(i) Find $I + 2J$. 1

(ii) Find $2I - J$. 1

(iii) Hence, or otherwise, find $\int \frac{\sin x}{\sin x + 2 \cos x} dx$. 2

Question 12 continues on Page 9

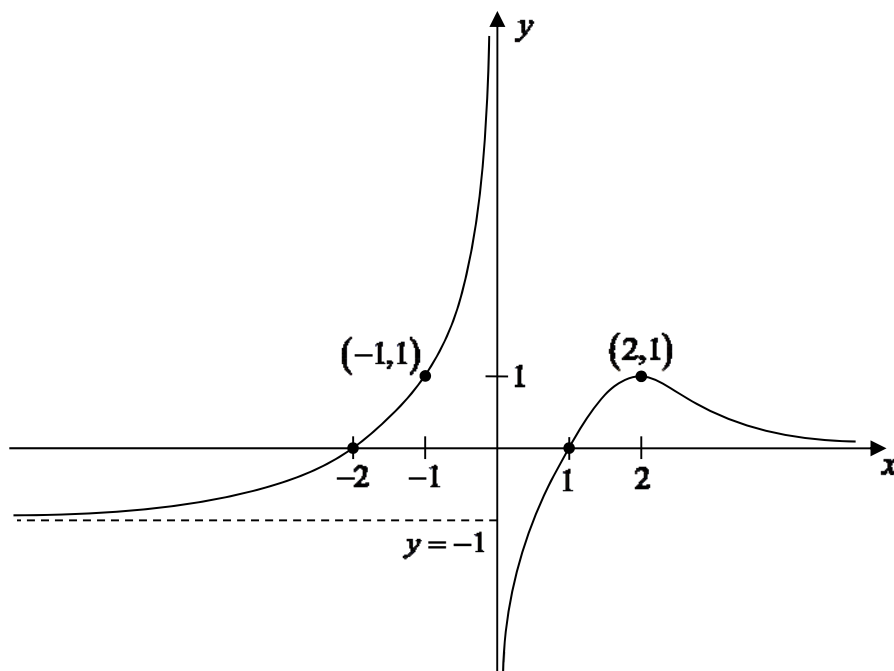
Question 12 (continued)

- (c) (i) Sketch the curve $f(x) = \frac{4x^2}{x^2 - 9}$ showing all intercepts and asymptotes. **2**
- (ii) Hence sketch $|y| = f(x)$ on a separate number plane. **2**
- (d) A relation is defined by the equation $\tan^{-1}(x^2) + \tan^{-1}(y^2) = \frac{\pi}{4}$.
- (i) Find $\frac{dy}{dx}$ in terms of x and y . **1**
- (ii) Find the gradient of the curve at the point where $x = \frac{1}{\sqrt{2}}$ and $y < 0$. **2**

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows a sketch of $y = f(x)$ with asymptotes at $x = 0$, $y = -1$ and $y = 0$. There is a maximum turning point at $(2, 1)$ and the curve passes through $(-1, 1)$.



Neatly sketch the graphs of the following showing all important information, including the coordinates of any new points which can be determined.

- | | | |
|------|---|----------|
| (i) | $y^2 = f(x)$ | 2 |
| (ii) | $y = e^{f(x)}$ | 2 |
| | | |
| (b) | (i) Prove that for any polynomial $P(x)$, if k is a zero of multiplicity r , then k is a zero of multiplicity $r - 1$ of $P'(x)$. | 1 |
| | | |
| (ii) | Given that the polynomial $P(x) = x^4 + 5x^3 + 9x^2 + 7x + 2$ has a zero of multiplicity 3, factorise $P(x)$. | 3 |

Question 13 continues on Page 11

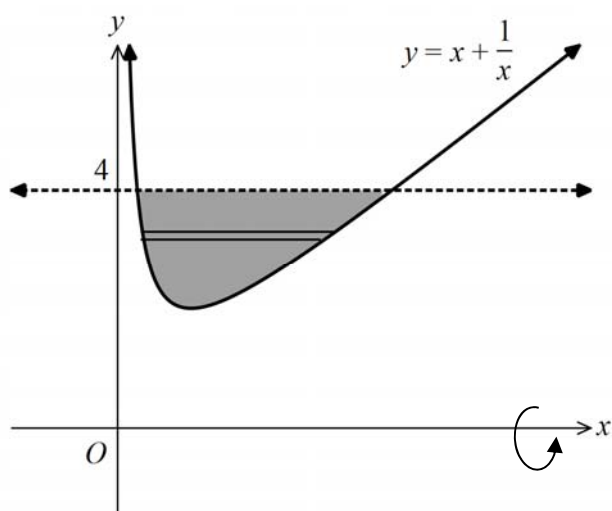
Question 13 (continued)

- (c) The cubic equation $x^3 + 4x + 3 = 0$ has roots α , β and γ .
- (i) Find a polynomial equation whose roots are α^2 , β^2 and γ^2 . **2**
- (ii) Hence, or otherwise, find the value of $\alpha^4 + \beta^4 + \gamma^4$. **2**
- (d) A sequence is defined by $a_1 = 1$, $a_2 = 8$ and $a_{n+2} = a_{n+1} + 2a_n$ for all positive integers n . Use Mathematical Induction to prove that $a_n = 3 \times 2^{n-1} + 2(-1)^n$. **3**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Let x be a positive real number. Show that $x + \frac{1}{x} \geq 2$. 1
- (ii) The region bounded by the curve $y = x + \frac{1}{x}$ and the line $y = 4$ is rotated about the x -axis. 4



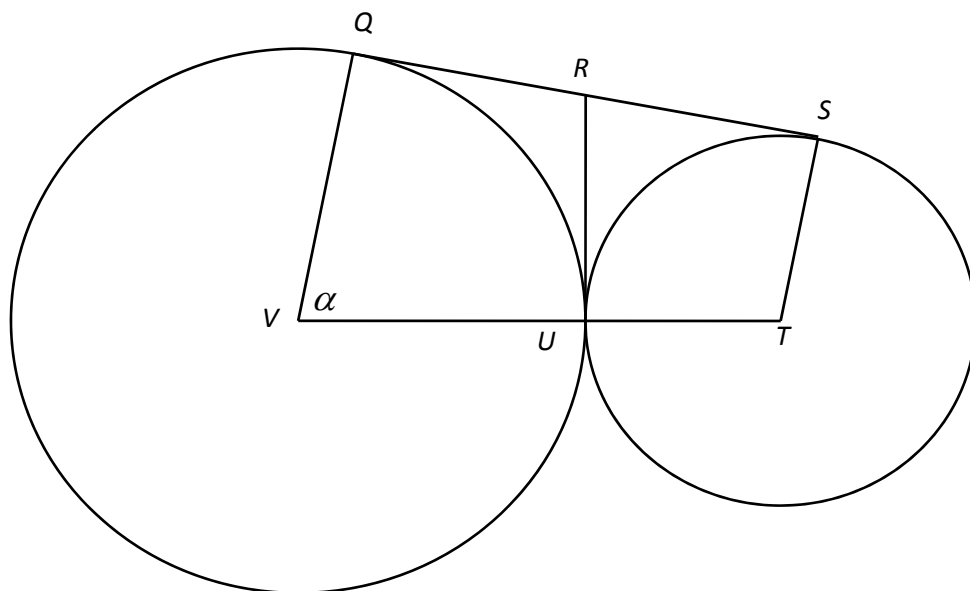
Use the method of cylindrical shells to show that the volume of the solid of revolution formed is $16\pi\sqrt{3}$ units³.

- (b) (i) Show that if x and y are positive and $x^3 + x^2 = y^3 - y^2$, then $x < y$. 2
- (ii) Show that if $0 < x \leq y - 1$, then $x^3 + x^2 < y^3 - y^2$. 2

Question 14 continues on Page 13

Question 14 (continued)

- (c) In the diagram, VUT is a straight line joining V and T , the centres of the circles. QS and RU are common tangents. Let $\angle QVU = \alpha$.



Copy the diagram into your answer booklet.

- | | | |
|-------|---|----------|
| (i) | Explain why $QRUV$ and $RSTU$ are cyclic quadrilaterals. | 1 |
| (ii) | Show that $\triangle SRU$ is similar to $\triangle QVU$. | 3 |
| (iii) | Show that QU is parallel to RT . | 2 |

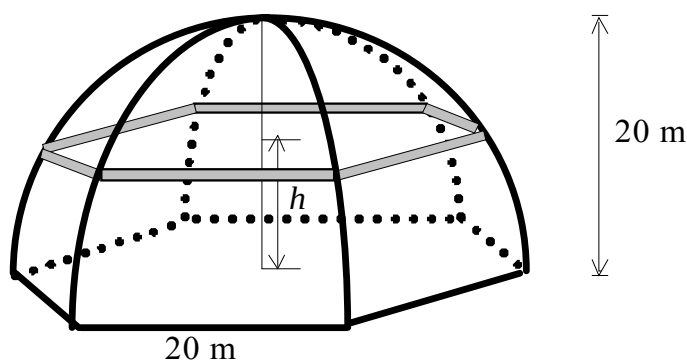
End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Determine $\int \cos^2 x \sin^7 x \, dx$.

3

- (b) A dome is sitting on a regular hexagonal base of side 20 metres. The height of the dome is also 20 metres. Each strut of the dome is a quarter of a circle with its centre at the centre of the hexagonal base.



A cross-sectional slice is taken parallel to the base of the dome.

- (i) If the slice is h metres above the base, deduce that the length of each side is $\sqrt{400 - h^2}$.

2

- (ii) Show that the area of the cross-section is $A = \frac{3\sqrt{3}}{2}(400 - h^2)$.

1

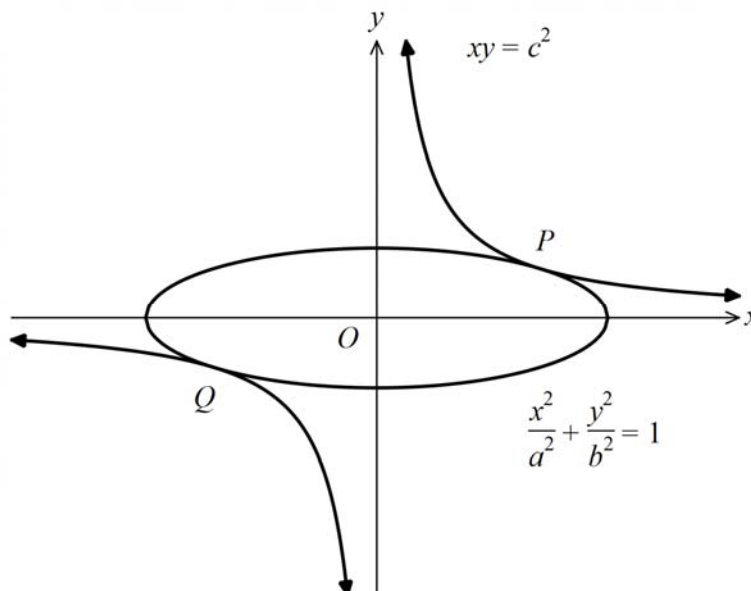
- (iii) Find the volume of the solid.

2

Question 15 continues on Page 15

Question 15 (continued)

- (c) The rectangular hyperbola $x = ct, y = \frac{c}{t}$, where $c > 0$, touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, at points P and Q , where $P\left(cp, \frac{c}{p}\right)$ lies in the first quadrant.



- (i) Explain why the equation $(bc)^2 t^4 - (ab)^2 t^2 + (ca)^2 = 0$ has roots $p, p, -p, -p$ where $p > 0$. 2
- (ii) Deduce that $p = \frac{a}{c\sqrt{2}}$ and $ab = 2c^2$. 2
- (iii) Show that if S and S' are the foci of the hyperbola $xy = c^2$, then the quadrilateral with vertices P, S, Q and S' has area $2c(a - b)$. 3

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Consider $I_n = \int \frac{x^n}{\sqrt{a^2 - x^2}} dx, \quad n \geq 0.$

(i) Show that $nI_n = -x^{n-1}\sqrt{a^2 - x^2} + a^2(n-1)I_{n-2}$ where $n \geq 2$. **3**

(ii) Hence find $\int \frac{x^2}{\sqrt{16 - x^2}} dx$. **1**

(b) Let $P(x)$ be a polynomial of degree n , where n is odd.

It is known that $P(k) = \frac{k}{k+1}$ for $k = 0, 1, 2, \dots, n$.

(i) $Q(x)$ is a polynomial such that $Q(x) = (x+1)P(x) - x$. Show that the zeroes of $Q(x)$ are $x = 0, 1, 2, \dots, n$. **1**

(ii) Let A be the leading coefficient of $Q(x)$. Factor $Q(x)$, **2**
and show that $A = \frac{1}{1 \times 2 \times 3 \times \dots \times n \times (n+1)} = \frac{1}{(n+1)!}$.

(iii) Find $P(n+1)$. **1**

Question 16 continues on Page 17

Question 16 (continued)

- (c) (i) Show that $x - \log_e(1+x) > 0$ for $x > 0$. **2**
- (ii) Hence show that $\sum_{k=1}^n \frac{1}{k} > \log_e(n+1)$. **2**
- (iii) Hence by considering $x + \log_e(1-x)$, show that $\sum_{k=1}^{\infty} \frac{1}{k^2} < 1 + \log_e 2$. **3**

End of paper



NORTH SYDNEY GIRLS HIGH SCHOOL
2019 HSC Trial Examination

Mathematics Extension 2- Solutions

1 What is the value of $(2+i^5)(i^2-i^3)$?

(A) $5-5i$

(B) $7-i$

(C) $-1+3i$

☒ (D) $-3+i$

$$\begin{aligned} & (2+i^5)i^2(1-i) \\ &= -(2+i)(1-i) \\ &= -2-i+2i-1 \\ &= -3+i \end{aligned}$$

2 Which expression is equal to $\int x^2 \cos x \, dx$?

(A) $x^2 \sin x + \int 2x \sin x \, dx$

☒ (B) $x^2 \sin x - \int 2x \sin x \, dx$

(C) $2x \sin x - \int x^2 \sin x \, dx$

(D) $2x \sin x + \int x^2 \sin x \, dx$

$$\begin{aligned} u &= x^2 & v' &= \cos x \\ u' &= 2x & v &= \sin x \\ \int x^2 \cos x \, dx &= x^2 \sin x - \int 2x \sin x \, dx \end{aligned}$$

3 A directrix of an ellipse has the equation $x = \frac{25}{4}$ and one of its foci has the coordinates $(-4, 0)$. What is the equation of the ellipse?

(A) $\frac{x^2}{5} + \frac{y^2}{3} = 1$

(B) $\frac{x^2}{3} + \frac{y^2}{5} = 1$

☒ (C) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

(D) $\frac{x^2}{9} + \frac{y^2}{25} = 1$

$$\begin{aligned} \frac{a}{e} &= \frac{25}{4}, & ae &= 4 \\ a \times \frac{4}{a} &= 25 \\ a^2 &= 25 \\ \text{Foci on the } x\text{-axis} & \therefore C \end{aligned}$$

- 4 Given the substitution $x = \pi - y$, which of the following is equal to $\int_0^{\pi} x \sin x \, dx$?

(A) $\int_{-\pi}^{\pi} \sin x \, dx$

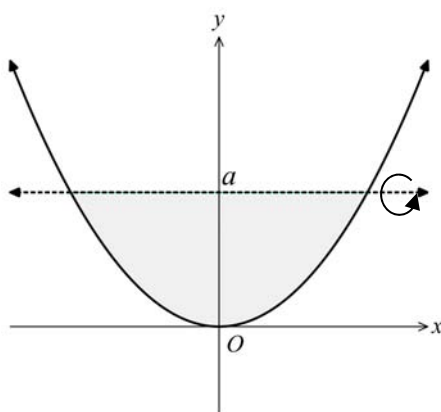
(B) $\frac{\pi}{2} \int_0^{\pi} \sin x \, dx$

(C) $\pi \int_0^{\pi} \sin x \, dx$

(D) $\int_0^{\pi} \sin x \, dx$

$$\begin{aligned} \int_0^{\pi} x \sin x \, dx &= \int_{\pi}^0 -(\pi - y) \sin(\pi - y) \, dy \quad \left[\begin{array}{l} \text{Let } x = \pi - y \\ dx = -dy \\ \text{when } x=0, y=\pi \\ x=\pi, y=0 \end{array} \right] \\ &= \int_0^{\pi} (\pi - y) \sin y \, dy \\ &= \int_0^{\pi} \pi \sin y \, dy - \int_0^{\pi} y \sin y \, dy \\ 2 \int_0^{\pi} x \sin x \, dx &= \int_0^{\pi} \pi \sin y \, dy \\ \therefore \int_0^{\pi} x \sin x \, dx &= \frac{\pi}{2} \int_0^{\pi} \sin y \, dy \end{aligned}$$

- 5 The region bounded by the parabola $x^2 = 4ay$ and the line $y = a$ is rotated about the line $y = a$ to form a solid. Which expression represents the volume of the solid?



(A) $2\pi \int_0^{2a} \left(a - \frac{x^2}{4a} \right)^2 dx$

(B) $2\pi \int_0^{2a} \left(a^2 - \left(\frac{x^2}{4a} \right)^2 \right) dx$

(C) $\pi \int_0^{2a} \left(a - \frac{x^2}{4a} \right)^2 dx$

(D) $\pi \int_0^{2a} \left(a^2 - \left(\frac{x^2}{4a} \right)^2 \right) dx$

$$\begin{aligned} \Delta A &= \pi (a - y)^2 & y &= \frac{x^2}{4a} & y &= a \\ &= \pi \left(a - \frac{x^2}{4a} \right)^2 & \frac{x^2}{4a} &= a \\ V &= \pi \int_{-2a}^{2a} \left(a - \frac{x^2}{4a} \right)^2 dx & x^2 &= 4a^2 \\ &= 2\pi \int_0^{2a} \left(a - \frac{x^2}{4a} \right)^2 dx & x &= \pm 2a \end{aligned}$$

- 6 Let e be the eccentricity of a conic, centred at the origin, with both foci on the x -axis. Which of the following is NOT true?

(A) If two ellipses have the same foci and directrices, then they have the same equation.

since $b^2 = a^2(1 - e^2)$, if a and e are the same then b is also the same.

(B) If two hyperbolae have equal eccentricity, then they share the same asymptotes.

since $\frac{b^2}{a^2} = e^2 - 1$, if e is the same then $\pm \frac{b}{a}$ is the same.

☒ (C) For the hyperbola, as $e \rightarrow \infty$, the asymptotes approach the x -axis.

since $\frac{b^2}{a^2} = e^2 - 1$, as $e \rightarrow \infty$ $\frac{b}{a} \rightarrow \infty$
 $\therefore$ asymptotes do not approach the x -axis.

(D) For the ellipse, as $e \rightarrow 0$, the directrices move further away from the origin whilst the foci approach the origin.

As $e \rightarrow 0$, $\frac{a}{e} \rightarrow \infty$ and $ae \rightarrow 0$.

- 7 Which complex number is a root of $z^6 + i = 0$?

(A) $-1 - i$

(B) $-1 + i$

(C) $-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$

☒ (D) $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$

$$\begin{aligned} z^6 &= \text{cis}\left(-\frac{\pi}{2} + 2k\pi\right) \\ z &= \text{cis}\left(\frac{4k-1}{12}\pi\right) \quad \text{By De Moivre's Theorem} \\ &= \text{cis}\left(-\frac{9\pi}{12}\right), \text{cis}\left(-\frac{5\pi}{12}\right), \text{cis}\left(-\frac{\pi}{12}\right), \text{cis}\left(\frac{3\pi}{12}\right), \text{cis}\left(\frac{7\pi}{12}\right), \text{cis}\left(\frac{11\pi}{12}\right) \\ \text{Now } \text{cis}\left(-\frac{9\pi}{12}\right) &= \text{cis}\left(-\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \end{aligned}$$

- 8 If the tangent at the point $(2 \sec \phi, 3 \tan \phi)$ on the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$ is parallel to $3x - y + 4 = 0$, then what is a possible value of ϕ ?

(A) 45°

(B) 60°

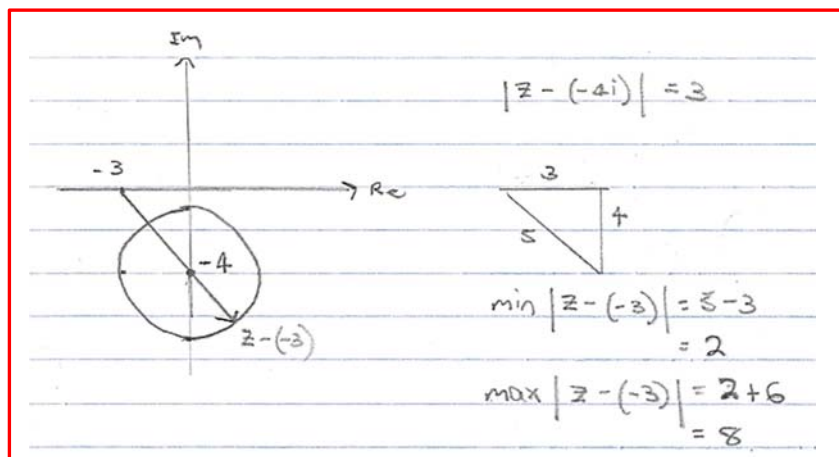
☒ (C) 30°

(D) 75°

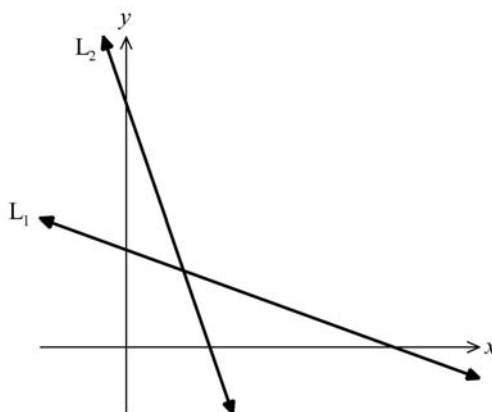
$$\begin{aligned} \frac{dx}{d\phi} &= 2 \sec \phi \tan \phi & \frac{dy}{d\phi} &= 3 \sec^2 \phi \\ \frac{dy}{dx} &= 3 \sec^2 \phi \times \frac{1}{2 \sec \phi \tan \phi} \\ &= \frac{3}{2 \sin \phi} \\ \frac{3}{2 \sin \phi} &= 3 \Rightarrow \sin \phi = \frac{1}{2} \\ & \phi = 30^\circ \end{aligned}$$

- 9 Let the complex number z satisfy the equation $|z + 4i| = 3$. What are the greatest and least values of $|z + 3|$?

- (A) 8 and 2
(B) 5 and 2
(C) 8 and 3
(D) 8 and 5



- 10 The diagram below shows the graphs of the straight lines L_1 and L_2 , whose equations are $y = ax + b$ and $y = cx + d$ respectively.

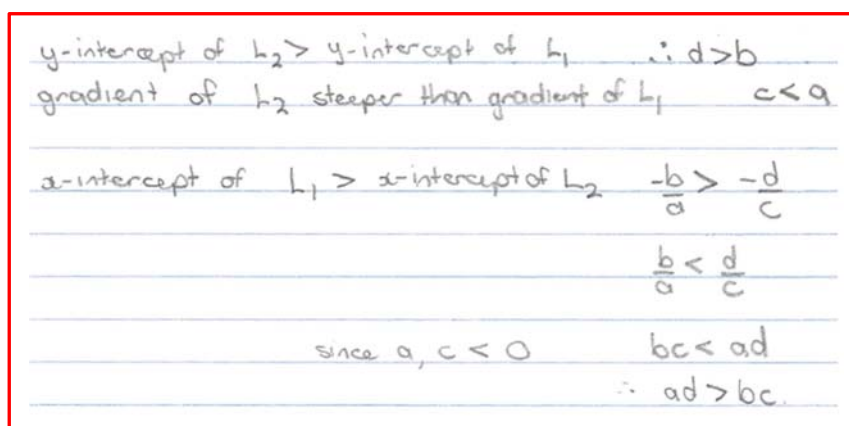


Which of the following are true?

- I. $c < a$
II. $d > b$
III. $ad > bc$

- (A) I and II only
(B) I and III only
(C) II and III only

- (D) I, II and III



Question 11

- (a) Express $\frac{4+3i}{2-i}$ in the form $x+iy$, where x and y are real.

2

$$\begin{aligned}\frac{4+3i}{2-i} \times \frac{2+i}{2+i} &= \frac{8+4i+6i-3}{4+1} \\ &= \frac{5+10i}{5} \\ &= 1+2i\end{aligned}$$

- (b) Consider the complex numbers $z = -1 + \sqrt{3}i$ and $w = 2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$.

- (i) Express z in modulus-argument form.

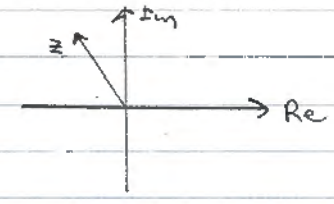
2

- (ii) Find the argument of $\frac{z}{w}$.

1

$$\begin{aligned}\text{i) } |z| &= \sqrt{(-1)^2 + (\sqrt{3})^2} \\ &= 2\end{aligned}$$

$$\tan(\arg(z)) = -\sqrt{3}$$

$$\arg(z) = \frac{2\pi}{3}$$


$$\begin{aligned}z &= 2\text{cis}\frac{2\pi}{3} \\ &= 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)\end{aligned}$$

$$\begin{aligned}\text{ii) } \arg\left(\frac{z}{w}\right) &= \frac{2\pi}{3} - \frac{\pi}{5} \\ &= \frac{7\pi}{15}\end{aligned}$$

- (c) (i) Express $\frac{3x^2+x+6}{(x^2+9)(x-1)}$ in the form $\frac{Ax+B}{x^2+9} + \frac{C}{x-1}$.

2

- (ii) Hence find $\int \frac{3x^2+x+6}{(x^2+9)(x-1)} dx$.

2

$$\begin{aligned}\text{i) } \frac{3x^2+x+6}{(x^2+9)(x-1)} &= \frac{Ax+B}{x^2+9} + \frac{C}{x-1} \\ &= \frac{(Ax+B)(x-1) + C(x^2+9)}{(x^2+9)(x-1)} \\ 3x^2+x+6 &= (Ax+B)(x-1) + C(x^2+9)\end{aligned}$$

$$\text{sub } x=1, \quad 10 = 10C \\ C = 1$$

$$\text{equate coefficients of } x^2, \quad 3 = A + C \Rightarrow A = 2.$$

$$\text{sub } x=0, \quad 6 = -B + 9C \\ -3 = -B \Rightarrow B = 3$$

$$\frac{3x^2 + x + 6}{(x^2 + 9)(x-1)} = \frac{2x+3}{x^2+9} + \frac{1}{x-1}$$

$$\begin{aligned} \text{ii) } \int \frac{3x^2 + x + 6}{(x^2 + 9)(x-1)} dx &= \int \frac{2x}{x^2+9} dx + \int \frac{3}{x^2+9} dx + \int \frac{dx}{x-1} \\ &= \log_e(x^2+9) + \tan^{-1} \frac{x}{3} + \log_e|x-1| + C \end{aligned}$$

- (d) The equation $|z-2| - |z+2| = \pm 2$ corresponds to a conic in the Argand diagram. Sketch the conic, showing any asymptotes, foci and directrices.

3

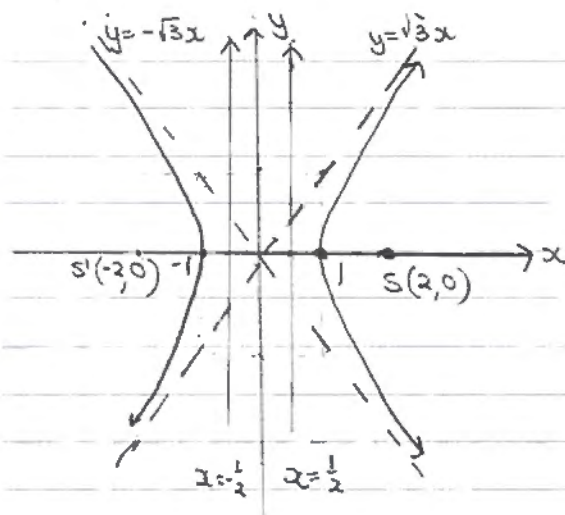
$$\begin{aligned} \text{foci at } (\pm 2, 0) & \text{ since } PS - PS' = \pm 2a \\ a &= 1 \\ ae &= 2 \\ \therefore e &= 2 \end{aligned} \quad \text{(alternate locus definition of hyperbola)}$$

$$\begin{aligned} b^2 &= a^2(e^2 - 1) \\ b^2 &= 4 - 1 \Rightarrow b = \sqrt{3} \end{aligned}$$

$$\text{directrices: } x = \pm \frac{a}{e}$$

$$x = \pm \frac{1}{2}$$

$$\text{asymptotes: } y = \pm \sqrt{3}x$$



(e) The polynomial $P(x) = x^4 + 3x^3 - x^2 - 13x - 10$ has a zero at $x = -2 - i$.

(i) Explain why $x = -2 + i$ is also a zero.

1

(ii) Fully factorise $P(x)$ over the real numbers.

2

i) complex roots of polynomial with real coefficients occur in conjugate pairs.

ii) Let the roots be $\alpha, \beta, -2 \pm i$

$$\text{sum of roots: } \alpha + \beta - 4 = -3$$

$$\alpha + \beta = 1$$

$$\text{product of roots: } \alpha\beta = -10$$

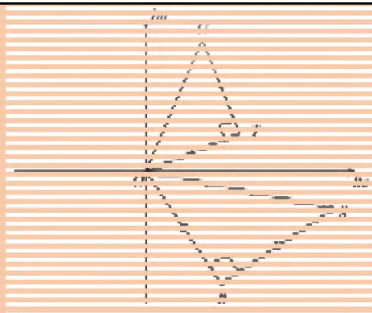
$$\alpha\beta = -2$$

The equation $x^2 - x - 2 = 0$ has roots α, β .

$$\begin{aligned} P(x) &= (x - (-2+i))(x - (-2-i))(x^2 - x - 2) \\ &= (x^2 + 4x + 5)(x - 2)(x + 1) \text{ over } \mathbb{R} \end{aligned}$$

Question 12

(a)



In the Argand diagram above, the points O, R, S, T and U correspond to the complex numbers $0, r, s, t$ and u respectively. The triangles ORS and OTU are right-angled isosceles triangles. Let $\omega = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$.

- (i) Explain why $u = \sqrt{2} \omega t$. 1
- (ii) Show that $r = \frac{s}{\sqrt{2} \omega}$. 1
- (iii) Using complex numbers show that $\frac{SU}{RT} = \sqrt{2}$. 2

i) To transform $\vec{OT}$ to $\vec{OU}$, rotate anti-clockwise by $\frac{\pi}{4}$ and scale by $\sqrt{2}$.

(Right-angled isosceles Δ lengths in ratio $1:1:\sqrt{2}$)
i.e. multiply by $\sqrt{2} \text{cis} \frac{\pi}{4} = \sqrt{2} \omega$

ii) To transform $\vec{OS}$ to $\vec{OR}$, rotate clockwise by $\frac{\pi}{4}$ and scale by $\frac{1}{\sqrt{2}}$.

i.e. multiply by $\frac{1}{\sqrt{2}} \text{cis}(-\frac{\pi}{4}) = \frac{1}{\sqrt{2}} \omega^{-1}$

$$\begin{aligned} \text{iii) } \vec{SU} &= \vec{OU} - \vec{OS} \\ &= u - s \\ &= \sqrt{2} \omega t - s \end{aligned}$$

$$\begin{aligned} \vec{RT} &= \vec{OT} - \vec{OR} \\ &= t - r \\ &= t - \frac{s}{\sqrt{2} \omega} \end{aligned}$$

$$\frac{|\vec{SU}|}{|\vec{RT}|} = \frac{|\sqrt{2} \omega t - s|}{|t - \frac{s}{\sqrt{2} \omega}|} = \frac{|\sqrt{2} \omega t - s|}{\frac{|\sqrt{2} \omega t - s|}{\sqrt{2} \omega}}$$

$$= \frac{|\sqrt{2} \omega t - s| \times |\sqrt{2} \omega|}{|\sqrt{2} \omega t - s|}$$

$$\begin{aligned} &= |\sqrt{2} \omega| \\ &= \sqrt{2} \end{aligned}$$

(b) Let $I = \int \frac{\sin x}{\sin x + 2 \cos x} dx$ and $J = \int \frac{\cos x}{\sin x + 2 \cos x} dx$.

(i) Find $I + 2J$.

1

(ii) Find $2I - J$.

1

(iii) Hence, or otherwise, find $\int \frac{\sin x}{\sin x + 2 \cos x} dx$.

2

i) $I + 2J = \int \frac{\sin x + 2 \cos x}{\sin x + 2 \cos x} dx$

$$= \int dx$$

$$= x + C$$

ii) $2I - J = \int \frac{2 \sin x - \cos x}{\sin x + 2 \cos x} dx$

$$= - \int \frac{\cos x - 2 \sin x}{\sin x + 2 \cos x} dx$$

$$= - \log_e |\sin x + 2 \cos x| + C_1$$

iii) $I + J + 2(2I - J) = x - 2 \log_e |\sin x + 2 \cos x| + C_2$

$$5I = x - 2 \log_e |\sin x + 2 \cos x| + C_2$$

$$I = \frac{x}{5} - \frac{2}{5} \log_e |\sin x + 2 \cos x| + C_3$$

(c) (i) Sketch the curve $f(x) = \frac{4x^2}{x^2 - 9}$ showing all intercepts and asymptotes.

2

(ii) Hence sketch $|y| = f(x)$ on a separate number plane.

2

i) vertical asymptotes: $x = \pm 3$

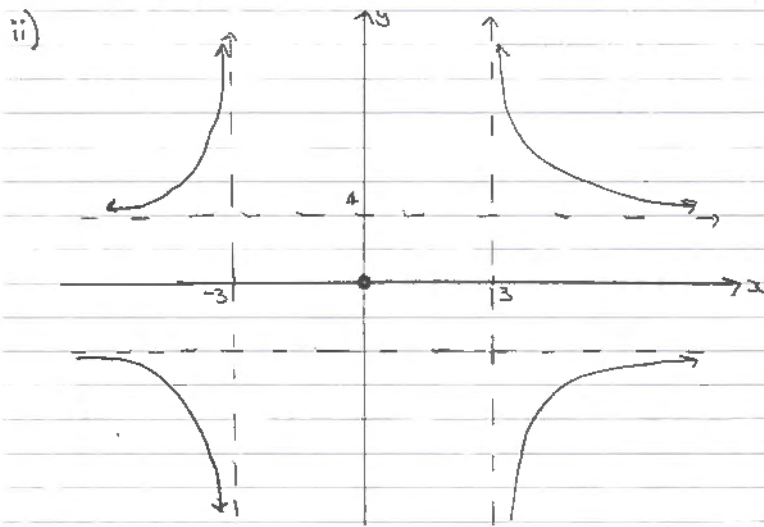
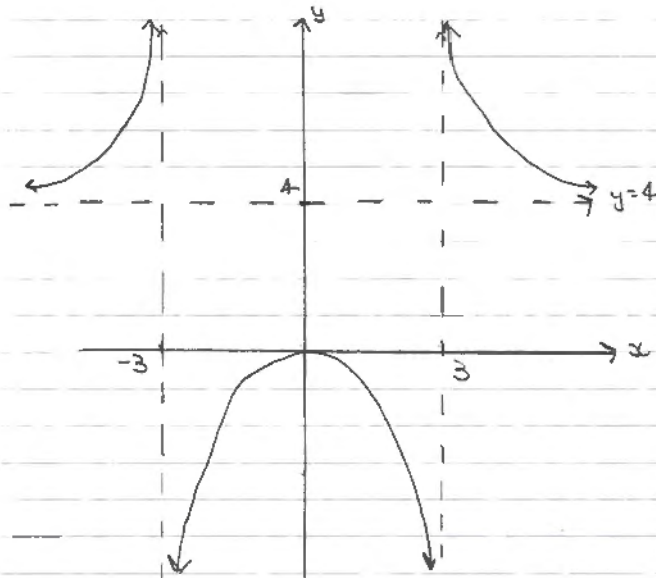
horizontal asymptote: $y = 4$

$f(-x) = f(x)$, $\therefore$ even function

when $x = 0$, $y = 0$

crosses horizontal asymptote if $4 = \frac{4x^2}{x^2 - 9}$

$$4x^2 - 36 = 4x^2, \text{ No soln.}$$



(d) A relation is defined by the equation $\tan^{-1}(x^2) + \tan^{-1}(y^2) = \frac{\pi}{4}$.

(i) Find $\frac{dy}{dx}$ in terms of x and y . 1

(ii) Find the gradient of the curve at the point where $x = \frac{1}{\sqrt{2}}$ and $y < 0$. 2

i) $\tan^{-1}(x^2) + \tan^{-1}(y^2) = \frac{\pi}{4}$

differentiating implicitly,

$$\frac{2x}{1+x^2} + \frac{2y}{1+y^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-2x}{1+x^2} \times \frac{1+y^2}{2y}$$

$$= \frac{-x(1+y^2)}{y(1+x^2)}$$

$$\text{ii) when } x = \frac{1}{\sqrt{2}}, \tan^{-1}\left(\frac{1}{\sqrt{2}}\right) + \tan^{-1}(y^2) = \frac{\pi}{4}$$

$$\tan^{-1}(y^2) = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$y^2 = \tan\left(\frac{\pi}{4} - \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)\right)$$

$$= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}$$

$$= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}$$

$$= \frac{1}{3}$$

$$y = -\frac{1}{\sqrt{3}} \quad (y < 0)$$

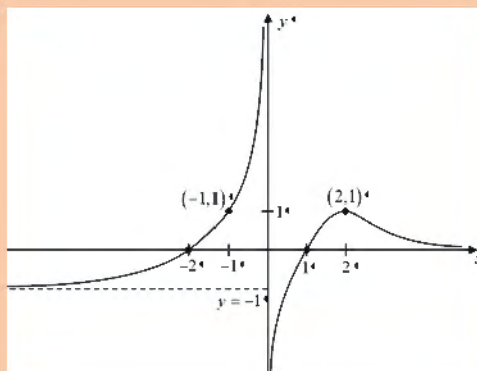
$$\therefore \frac{dy}{dx} = \frac{-\frac{1}{\sqrt{2}} \left(1 + \frac{1}{9}\right)}{-\frac{1}{\sqrt{3}} \left(1 + \frac{1}{4}\right)} = \frac{\sqrt{3} \times \frac{10}{9}}{\sqrt{2} \times \frac{5}{4}}$$

$$= \frac{10}{3\sqrt{3}} \times \frac{2\sqrt{2}}{5}$$

$$= \frac{4\sqrt{2}}{3\sqrt{3}} \quad \text{or} \quad \frac{8\sqrt{3}}{9\sqrt{2}}$$

Question 13

- (a) The diagram shows a sketch of $y = f(x)$ with asymptotes at $x = 0$, $y = -1$ and $y = 0$. There is a maximum turning point at $(2, 1)$ and the curve passes through $(-1, 1)$.



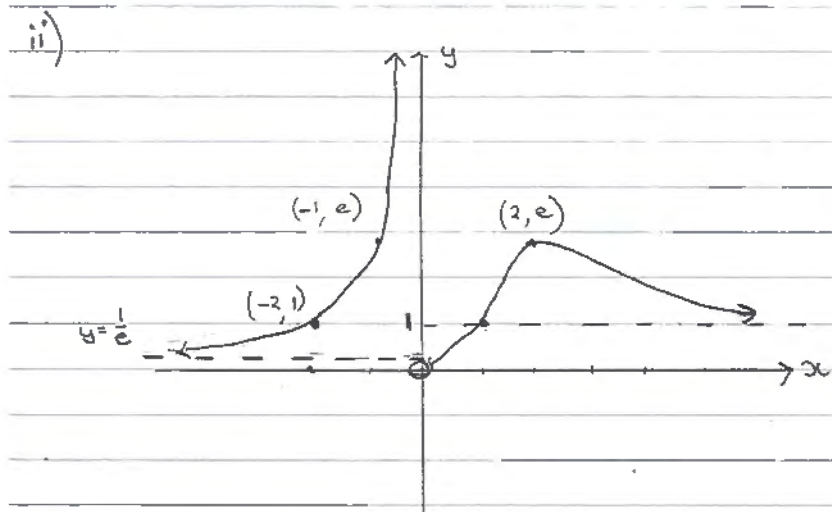
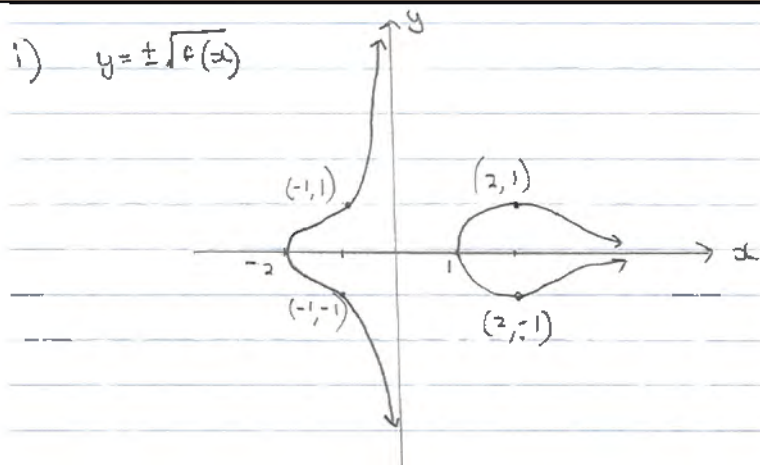
Neatly sketch the graphs of the following showing all important information, including the coordinates of any new points which can be determined.

(i) $y^2 = f(x)$

2

(ii) $y = e^{f(x)}$

2



- | | | | |
|-----|------|---|---|
| (b) | (i) | Prove that for any polynomial $P(x)$, if k is a zero of multiplicity r , then k is a zero of multiplicity $r-1$ of $P'(x)$. | 1 |
| | (ii) | Given that the polynomial $P(x) = x^4 + 5x^3 + 9x^2 + 7x + 2$ has a zero of multiplicity 3, factorise $P(x)$. | 3 |

i) Let $P(x) = (x-k)^r Q(x)$ where $Q(k) \neq 0$
 $P'(x) = r(x-k)^{r-1} Q(x) + (x-k)^r \cdot Q'(x)$

$$= (x-k)^{r-1} [rQ(x) + (x-k)Q'(x)]$$

$$= (x-k)^{r-1} [R(x)] \quad \text{where } R(k) \neq 0$$

$\therefore (x-k)^{r-1}$ is a zero of $P'(x)$ with multiplicity $(r-1)$.

ii) $P(x) = x^4 + 5x^3 + 9x^2 + 7x + 2$

$$P'(x) = 4x^3 + 15x^2 + 18x + 7$$

$$P''(x) = 12x^2 + 30x + 18$$

set $P''(x) = 0$, $12x^2 + 30x + 18 = 0$

$$2x^2 + 5x + 3 = 0$$

$$(2x+3)(x+1) = 0$$

$$x = -\frac{3}{2}, -1$$

$$P(-1) = 1 - 5 + 9 - 7 + 2$$

$$= 0$$

$\therefore -1$ is a triple root of $P(x) = 0$

Let the roots of $P(x) = 0$ be $-1, -1, -1, \alpha$

Sum of roots $\alpha - 3 = -5$

$$\alpha = -2$$

$$\therefore P(x) = (x+2)(x+1)^3$$

(c) The cubic equation $x^3 + 4x + 3 = 0$ has roots α , β and γ .

(i) Find a polynomial equation whose roots are α^2 , β^2 and γ^2 .

2

(ii) Hence, or otherwise, find the value of $\alpha^4 + \beta^4 + \gamma^4$.

2

i) Let $x^3 + 4x + 3 = 0$

Transforming the roots,

$$\alpha^2 = x$$

$$x = \sqrt{\alpha}$$

$$(\sqrt{\alpha})^3 + 4\sqrt{\alpha} + 3 = 0$$

$$\sqrt{\alpha}(\alpha + 4) = -3$$

$$\alpha(\alpha + 4)^2 = 9$$

$$\alpha^3 + 8\alpha^2 + 16\alpha - 9 = 0$$

This polynomial equation has roots α^2 , β^2 and γ^2

ii) Let $A = \alpha^2$, $B = \beta^2$, $C = \gamma^2$

$$\alpha^4 + \beta^4 + \gamma^4 = A^2 + B^2 + C^2$$

$$\begin{aligned} A^2 + B^2 + C^2 &= (A+B+C)^2 - 2(AB+AC+BC) \\ &= (-8)^2 - 2(16) \quad \text{From i)} \end{aligned}$$

$$= 32$$

(d) A sequence is defined by $a_1 = 1$, $a_2 = 8$ and $a_{n+2} = a_{n+1} + 2a_n$ for all positive integers n . Use Mathematical Induction to prove that $a_n = 3 \times 2^{n-1} + 2(-1)^n$.

3

Prove true for $n=1$ and $n=2$.

For $n=1$,

$$\text{LHS} = 1 \quad \text{RHS} = 3 \times 2^{1-1} + 2(-1)^1$$

$$= 1$$

for $n=2$,

$$\text{LHS} = 8 \quad \text{RHS} = 3 \times 2^{2-1} + 2(-1)^2$$

$$= 6 + 2$$

$$= 8$$

Assume true for $n=k$ and $n=k+1$

$$a_k = 3 \times 2^{k-1} + 2(-1)^k$$

$$a_{k+1} = 3 \times 2^k + 2(-1)^{k+1}$$

Prove true for $n = k+2$

$$\text{i.e., R.T.P. } a_{k+2} = 3 \times 2^{k+2-1} + 2(-1)^{k+2} \\ = 3 \times 2^{k+1} + 2(-1)^{k+2}$$

$$\text{LHS} = a_{k+2}$$

$$= a_{k+1} + 2a_k$$

$$= 3 \times 2^{k+1} + 2(-1)^{k+1} + 2[3 \times 2^{k+1} + 2(-1)^k] \quad (\text{By assumption})$$

$$= 3 \times 2^{k+1} + 2(-1)^{k+1} + 3 \times 2^{k+1} + 4(-1)^k$$

$$= 6 \times 2^{k+1} + 2(-1)(-1)^k + 4(-1)^k$$

$$= 3 \times 2^{k+1} + 2(-1)^k$$

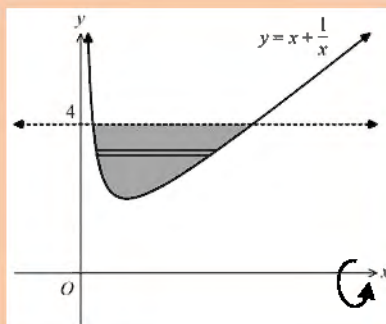
$$= 3 \times 2^{k+1} + 2(-1)^{k+2}$$

$$= \text{RHS.}$$

$\therefore$ the statement is true by Mathematical Induction.

Question 14 (15 marks)

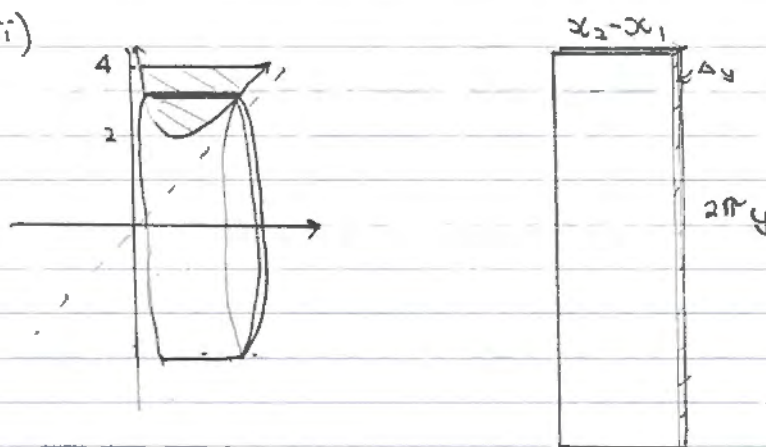
- (a) (i) Let x be a positive real number. Show that $x + \frac{1}{x} \geq 2$. 1
- (ii) The region bounded by the curve $y = x + \frac{1}{x}$ and the line $y = 4$ is rotated about the x -axis. 4



Use the method of cylindrical shells to show that the volume of the solid of revolution formed is $16\pi\sqrt{3}$ units³.

i) $(\sqrt{x} - \frac{1}{\sqrt{x}})^2 \geq 0$
 $x + \frac{1}{x} - 2 \geq 0$
 $x + \frac{1}{x} \geq 2$

ii)



$y = x + \frac{1}{x}$
 $xy = x^2 + 1$
 $x^2 - xy + 1 = 0$
 $x = \frac{-y \pm \sqrt{(-y)^2 - 4}}{2}$
 $= \frac{y \pm \sqrt{y^2 - 4}}{2}$
 $\therefore x_2 - x_1 = \frac{y + \sqrt{y^2 - 4}}{2} - \frac{y - \sqrt{y^2 - 4}}{2}$

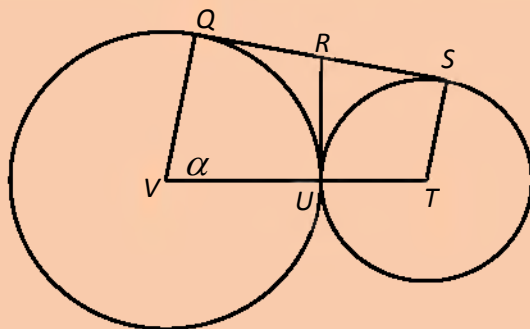
$$\begin{aligned}
 \Delta A &= \sqrt{y^2 - 4} \times 2\pi y \\
 \Delta V &= 2\pi y \sqrt{y^2 - 4} \Delta y \\
 V &= \lim_{\Delta y \rightarrow 0} \sum_{y=2}^4 2\pi y \sqrt{y^2 - 4} \Delta y \\
 &= \pi \int_2^4 2y \sqrt{y^2 - 4} dy \\
 &= \pi \left[\frac{2(y^2 - 4)^{3/2}}{3} \right]_2^4 \\
 &= \frac{2\pi}{3} (12^{3/2} - 0) \\
 &= \frac{2\pi}{3} (2\sqrt{3})^3 \\
 &= \frac{2\pi}{3} \times 8 \times 3\sqrt{3} \\
 &= 16\pi\sqrt{3} \text{ units}^3
 \end{aligned}$$

- | | | | |
|-----|------|--|---|
| (b) | (i) | Show that if x and y are positive and $x^3 + x^2 = y^3 - y^2$, then $x < y$. | 2 |
| | (ii) | Show that if $0 < x \leq y - 1$, then $x^3 + x^2 < y^3 - y^2$. | 2 |

$$\begin{aligned}
 \text{i)} \quad x^3 + x^2 &= y^3 - y^2 \\
 x^3 + x^2 &< y^3 & (x^2 > 0) \\
 x^3 &< y^3 & (y^2 > 0) \\
 x &< y & (x, y > 0)
 \end{aligned}$$

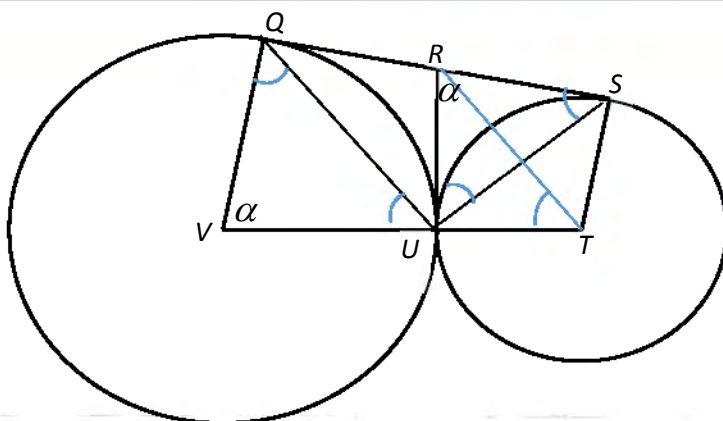
$$\begin{aligned}
 \text{ii)} \quad 0 < x &\leq (y-1) \\
 \text{squaring } 0 < x^2 &\leq y^2 - 2y + 1 \quad (1) \\
 \text{cubing } 0 < x^3 &\leq (y-1)^3 \\
 &= y^3 - 3y^2 + 3y - 1 \quad (2) \\
 (1) + (2) \quad x^3 + x^2 &\leq y^3 - 2y^2 + y \\
 &= y^3 - y^2 - (y^2 - y) \\
 &= y^3 - y^2 - y(y-1) \\
 &< y^3 - y^2 \quad y > 0 \text{ and } y-1 > 0
 \end{aligned}$$

- (c) In the diagram, VUT is a straight line joining V and T , the centres of the circles. QS and RU are common tangents. Let $\angle QVU = \alpha$.



Copy the diagram into your answer booklet.

- (i) Explain why $QRUV$ and $RSTU$ are cyclic quadrilaterals. 1
- (ii) Show that $\triangle SRU$ is similar to $\triangle QVU$. 3
- (iii) Show that QU is parallel to RT . 2



- i) In each case a pair of opposite angles are equal to 90° . (tangent perpendicular to radius)

$\therefore QRUV$ and $RSTU$ are cyclic quadrilaterals.
(opposite angles are supplementary.)

- ii) In $\triangle SRU$ and $\triangle QVU$

$$\angle QVU = \angle SRU \quad (\text{exterior angle of a cyclic quadrilateral})$$

$$= \alpha$$

$$QV = VU$$

$$\angle VQU = \angle QUV \quad (\text{angles opposite equal sides})$$

$$\angle QUV = \frac{180 - \alpha}{2}$$

$$RU = US \quad (\text{tangents from external point})$$

$$\therefore \angle RSU = \angle RUS \quad (\text{angles opposite equal sides})$$

$$\angle RSU = \frac{180 - \alpha}{2}$$

$$\therefore \angle VUQ = \angle RSU = \frac{180 - \alpha}{2}$$

$$\therefore \triangle SRU \parallel \triangle QVU \quad (\text{equiangular})$$

iii) In $RSTU$, $RS = RS$ (tangents from external point)

$$UT = TS \text{ (radii)}$$

$RSTU$ is a kite (two pairs of adjacent sides equal)

$$\angle UTS = 180 - \alpha \text{ (opposite angle of cyclic quad.)}$$

$$\angle RTU = \frac{180 - \alpha}{2} \text{ (diagonal bisects vertex angle)}$$

$$\therefore \angle QUV = \angle RTU = \frac{180 - \alpha}{2}$$

$QU \parallel RT$ (corresponding angles equal)

iii) ALTERNATE SOLUTION

$$\angle VUQ = \angle RSU \text{ (shown in part ii)}$$

construct RT

$$\angle RSU = \angle RTU$$

$$\therefore \angle RTU = \angle QUV$$

$QU \parallel RT$ (corresponding angles are equal)

Question 15

(a) Determine $\int \cos^2 x \sin^7 x \, dx$.

3

$$a) \quad \int \cos^2 x \sin^7 x \, dx = \int \cos^2 x (\sin^2 x)^3 \sin x \, dx$$

$$= \int \cos^2 x (1 - \cos^2 x)^3 \sin x \, dx$$

$$\text{Let } u = \cos x$$

$$-du = \sin x \, dx$$

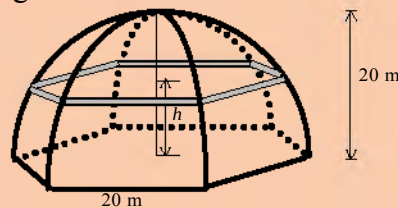
$$= - \int u^2 (1 - u^2)^3 \, du$$

$$= - \int (u^2 - 3u^4 + 3u^6 - u^8) \, du$$

$$= - \left(\frac{u^3}{3} - \frac{3u^5}{5} + \frac{3u^7}{7} - \frac{u^9}{9} \right) + C$$

$$= -\frac{1}{3} \cos^3 x + \frac{3}{5} \cos^5 x - \frac{3}{7} \cos^7 x + \frac{1}{9} \cos^9 x + C$$

- (b) A dome is sitting on a regular hexagonal base of side 20 metres. The height of the dome is also 20 metres. Each strut of the dome is a quarter of a circle with its centre at the centre of the hexagonal base.

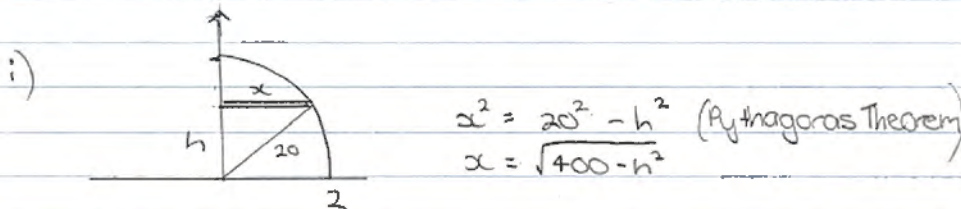


A cross-sectional slice is taken parallel to the base of the dome.

- (i) If the slice is h metres above the base, deduce that the length of each side is $\sqrt{400 - h^2}$. 2

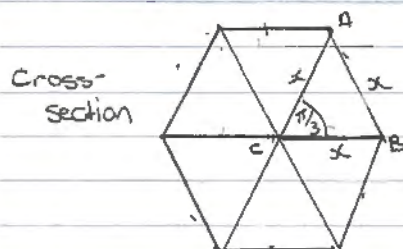
- (ii) Show that the area of the cross-section is $A = \frac{3\sqrt{3}}{2} (400 - h^2)$. 1

- (iii) Find the volume of the solid. 2



$$x^2 = 20^2 - h^2 \quad (\text{Pythagoras Theorem})$$

$$x = \sqrt{400 - h^2}$$



$\triangle ABC$ is equilateral triangle.
 $\therefore$ side length of hexagon is x .

ii) Area of hexagon:

$$A = 6 \times \frac{1}{2} \times a \times a \times \sin \frac{\pi}{3}$$

$$= 3a^2 \times \frac{\sqrt{3}}{2}$$

$$= \frac{3\sqrt{3}}{2} (400 - h^2)$$

iii) $\Delta V = A \Delta h$

$$V = \lim_{\Delta h \rightarrow 0} \sum_{h=0}^{20} \frac{3\sqrt{3}}{2} (400 - h^2) \Delta h$$

$$= \int_0^{20} \frac{3\sqrt{3}}{2} (400 - h^2) dh$$

$$= \frac{3\sqrt{3}}{2} \left[400h - \frac{h^3}{3} \right]_0^{20}$$

$$= \frac{3\sqrt{3}}{2} \left(8000 - \frac{8000}{3} \right)$$

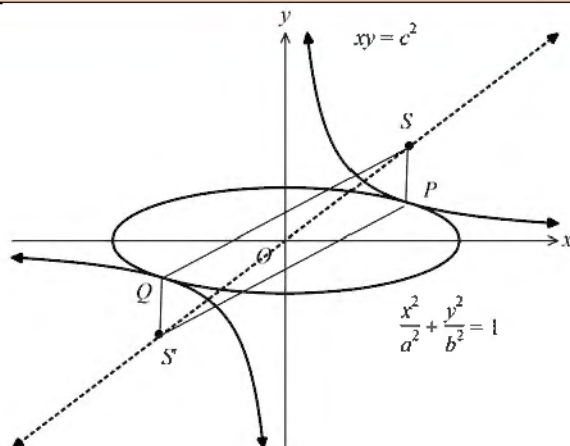
$$= 8000\sqrt{3} \text{ units}^3$$

(c) The rectangular hyperbola $x = ct$, $y = \frac{c}{t}$, where $c > 0$, touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, at points P and Q , where $P\left(cp, \frac{c}{p}\right)$ lies in the first quadrant.

(i) Explain why the equation $(bc)^2 t^4 - (ab)^2 t^2 + (ca)^2 = 0$ has roots $p, p, -p, -p$ where $p > 0$. 2

(ii) Deduce that $p = \frac{a}{c\sqrt{2}}$ and $ab = 2c^2$. 2

(iii) Show that if S and S' are the foci of the hyperbola $xy = c^2$, then the quadrilateral with vertices P, S, Q and S' has area $2c(a - b)$. 3



i) $x = ct$ $y = \frac{c}{t}$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

solving simultaneously for pts of intersection.

$$\frac{c^2 t^2}{a^2} + \frac{c^2}{b^2 t^2} = 1 \quad \times a^2 b^2 t^2$$

$$b^2 c^2 t^4 + a^2 c^2 = a^2 b^2 t^2$$

$$(bc)^2 t^4 - (ab)^2 t^2 + (ac)^2 = 0$$

This equation has 2 double roots as the ellipse and hyperbola are tangential to each other at the pts of contact.

Also by the symmetry of the ellipse & the hyperbola, P & Q are at opposite ends of a diameter, therefore their parameter values are opposite of each other.

Hence the roots are $p, p, -p, -p$ where p is the parameter value at P .

ii) Sum of roots in pairs:

$$(p)(p) + (p)(-p) + (p)(-p) + p(-p) + p(-p) + (-p)(-p)$$

$$\therefore -2p^2 = -\frac{(ab)^2}{(bc)^2}$$

$$p^2 = \frac{a^2}{2c^2}$$

$$\boxed{p = \frac{a}{c\sqrt{2}}} \quad (p > 0) \quad (1)$$

product of roots:

$$p \times p \times (-p) \times (-p) = \frac{(ca)^2}{(bc)^2}$$

$$p^4 = \frac{a^2}{b^2} \quad (2)$$

using ① & ②

$$p^4 = \frac{a^4}{c^4 \times 4} = \frac{a^2}{b^2} \div a^2$$

$$\frac{a^2}{4c^4} = \frac{1}{b^2}$$

$$a^2 b^2 = 4c^4$$

$$ab = 2c^2 \quad (ab > 0)$$

$$\therefore p = \frac{a}{c\sqrt{2}} \quad ab = 2c^2$$

iii) $SPS'Q$ is a parallelogram (by symmetry)

Area of $SPS'Q = 2 \times \text{Area } \triangle SPS'$

$$\text{Perpendicular distance from } P \text{ to } SS' = \left| cp - \frac{c}{p} \right|$$

$\downarrow$
 $x-y=0$ $\frac{\sqrt{1^2+1^2}}$

$$= \frac{\left| cp - \frac{c}{p} \right|}{\sqrt{2}}$$

$$SS' = 4c$$

$$\text{Area } \triangle SPS' = \frac{1}{2} \times \frac{1}{\sqrt{2}} \left| cp - \frac{c}{p} \right| \times 4c$$

$$= \sqrt{2} c^2 \left| p - \frac{1}{p} \right|$$

$$= \sqrt{2} c^2 \times \left(\frac{a}{c\sqrt{2}} - \frac{c\sqrt{2}}{a} \right) \quad \left(\begin{array}{l} \text{since } a > \frac{c}{\sqrt{2}} \\ \text{from diagram} \end{array} \right)$$

$$= \sqrt{2} c^2 \left(\frac{a^2 - 2c^2}{a\sqrt{2}} \right)$$

$$= \frac{c}{a} (a^2 - 2c^2)$$

$$= \frac{c}{a} (a^2 - ab) \quad \text{from ii)}$$

$$= c(a-b)$$

$$\therefore \text{Area } SPS'Q = 2 \times \text{Area } \triangle SPS'$$

$$= 2c(a-b)$$

Question 16.

(a) Consider $I_n = \int \frac{x^n}{\sqrt{a^2 - x^2}} dx$, $n \geq 0$.

(i) Show that $nI_n = -x^{n-1}\sqrt{a^2 - x^2} + a^2(n-1)I_{n-2}$ where $n \geq 2$. 3

(ii) Hence find $\int \frac{x^2}{\sqrt{16 - x^2}} dx$. 1

$$i) I_n = \int \frac{x^n}{\sqrt{a^2 - x^2}} dx \quad n \geq 0$$

$$u = x^{n-1} \quad v' = \frac{x}{\sqrt{a^2 - x^2}}$$

$$u' = (n-1)x^{n-2} \quad v = -\frac{2}{2} \sqrt{a^2 - x^2} \\ = -\sqrt{a^2 - x^2}$$

$$I_n = -x^{n-1}\sqrt{a^2 - x^2} + (n-1) \int x^{n-2} \sqrt{a^2 - x^2} dx \quad (\text{by parts})$$

$$= -x^{n-1}\sqrt{a^2 - x^2} + (n-1) \int x^{n-2} \frac{(a^2 - x^2)}{\sqrt{a^2 - x^2}} dx$$

$$= -x^{n-1}\sqrt{a^2 - x^2} + (n-1)a^2 \int \frac{x^{n-2}}{\sqrt{a^2 - x^2}} dx - (n-1) \int \frac{x^n}{\sqrt{a^2 - x^2}} dx$$

$$= -x^{n-1}\sqrt{a^2 - x^2} + (n-1)a^2 I_{n-2} - (n-1) I_n$$

$$ii) \int \frac{x^2}{\sqrt{16 - x^2}} dx = I_2 \quad (\text{where } a=4)$$

$$2I_2 = -x\sqrt{16 - x^2} + a^2(2-1)I_0 \quad (\text{from part i}) \\ = -x\sqrt{16 - x^2} + 16 \int \frac{1}{\sqrt{16 - x^2}} dx$$

$$I_2 = -\frac{x}{2}\sqrt{16 - x^2} + 8 \sin^{-1} \frac{x}{4} + C$$

(b) Let $P(x)$ be a polynomial of degree n , where n is odd.

It is known that $P(k) = \frac{k}{k+1}$ for $k = 0, 1, 2, \dots, n$.

(i) $Q(x)$ is a polynomial such that $Q(x) = (x+1)P(x) - x$. Show that the zeroes of $Q(x)$ are $x = 0, 1, 2, \dots, n$. 1

(ii) Let A be the leading coefficient of $Q(x)$. Factor $Q(x)$, and show that $A = \frac{1}{1 \times 2 \times 3 \times \dots \times n \times (n+1)} = \frac{1}{(n+1)!}$. 2

(iii) Find $P(n+1)$. 1

i) sub. $x = k$

$$Q(k) = (k+1)P(k) - k$$

$$= \cancel{(k+1)} \times \frac{k}{\cancel{k+1}} - k \quad \text{where } k = 0, 1, 2, \dots, n$$

$$= 0$$

$\therefore$ the zeroes of $Q(k)$ are $x = 0, 1, 2, \dots, n$
(by the factor theorem)

ii) $Q(x) = (x+1)P(x) - x$ ①

$$Q(x) = A x(x-1)(x-2) \dots (x-n) \quad \text{②}$$

sub $x = -1$ in ②

$$Q(-1) = -A(-2)(-3) \dots (-1-n)$$

$$= (-1)A(-2)(-3) \dots (-(n+1))$$

$$= (-1)^{n+1} A \times (n+1)!$$

$$= A \times (n+1)! \quad \left(\begin{array}{l} \text{since } n+1 \text{ is even,} \\ (-1)^{n+1} = 1 \end{array} \right)$$

sub $x = -1$ in ①

$$Q(-1) = (-1+1)P(-1) - (-1)$$

$$= 1$$

**

Equating * and *

$$\therefore A(n+1)! = 1$$

$$A = \frac{1}{(n+1)!}$$

iii) sub $x = n+1$

$$Q(n+1) = (n+1+1) \times P(n+1) - (n+1) \\ = (n+2) P(n+1) - (n+1) \quad \textcircled{A}$$

$$Q(n+1) = \frac{1}{(n+1)!} (n+1)(n+1-1)(n+1-2) \times \dots \times (n+1-n)$$

$$= \frac{1}{(n+1)!} (n+1)(n)(n-1) \times \dots \times (1)$$

$$= \frac{(n+1)!}{(n+1)!}$$

$$= 1 \quad \textcircled{B}$$

Equating $\textcircled{A} = \textcircled{B}$

$$(n+2)P(n+1) - (n+1) = 1$$

$$(n+2)P(n+1) = 1+n+1$$

$$P(n+1) = \frac{n+2}{n+2}$$

$$P(n+1) = 1$$

(c)	(i)	Show that $x - \log_e(1+x) > 0$ for $x > 0$.	2
	(ii)	Hence show that $\sum_{k=1}^n \frac{1}{k} > \log_e(n+1)$.	2
	(iii)	Hence by considering $x + \log_e(1-x)$, show that $\sum_{k=1}^{\infty} \frac{1}{k^2} < 1 + \log_e 2$.	3

i) $\frac{d}{dx} (x - \ln(1+x)) = 1 - \frac{1}{1+x}$

$$= \frac{1+x-1}{1+x}$$

$$= \frac{x}{1+x}$$

$$\geq 0 \quad \text{since } x > 0, 1+x > 0$$

$\therefore (x - \ln(1+x))$ is increasing for all $x > 0$.

$$\text{Also, if } x=0 \quad x - \ln(1+x) = 0$$

$$\therefore x - \log_e(1+x) > 0 \text{ for } x > 0.$$

ii) using part i)

$$\frac{1}{k} > \ln\left(1 + \frac{1}{k}\right) \quad \text{for all } k > 0$$

$$\frac{1}{k} > \ln\left(\frac{k+1}{k}\right)$$

$$\therefore \frac{1}{1} > \ln\left(\frac{2}{1}\right) \quad k=1$$

$$\frac{1}{2} > \ln\left(\frac{3}{2}\right) \quad k=2$$

...

$$\frac{1}{n} > \ln\left(\frac{n+1}{n}\right) \quad k=n.$$

Summing

$$\begin{aligned} 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} &> \ln\left(\frac{2}{1}\right) + \ln\left(\frac{3}{2}\right) + \dots + \ln\left(\frac{n+1}{n}\right) \\ &= \ln\left(\frac{2}{1} \times \frac{3}{2} \times \dots \times \frac{n+1}{n}\right) \\ &= \ln(n+1) \end{aligned}$$

$$\therefore \sum_{k=1}^n \frac{1}{k} > \ln(n+1)$$

$$\begin{aligned} \text{iii) } \frac{d}{d\alpha} (\alpha + \log_e(1-\alpha)) &= 1 - \frac{1}{1-\alpha} \\ &= \frac{1-\alpha-1}{1-\alpha} \\ &= \frac{\alpha}{\alpha-1} \end{aligned}$$

For $0 < \alpha < 1$, $\frac{\alpha}{\alpha-1} < 0$ [numerator positive
denominator negative]

$\therefore x + \log_e(1-x)$ is decreasing for $0 < x < 1$

Also, if $x=0$ $x + \log_e(1-x) = 0$

$\therefore x + \log_e(1-x) < 0$ for $0 < x < 1$

and so $\frac{1}{k^2} + \log_e\left(1 - \frac{1}{k^2}\right) < 0$ for all $k > 1$

$$\therefore \sum_{k=2}^n \frac{1}{k^2} < \sum_{k=2}^n -\ln\left(1 - \frac{1}{k^2}\right)$$

$$= \sum_{k=2}^n -\ln\left(\frac{k^2-1}{k^2}\right)$$

$$= \sum_{k=2}^n -\ln\left(\frac{(k-1)(k+1)}{k^2}\right)$$

$$= \sum_{k=2}^n \ln\left(\frac{k^2}{(k-1)(k+1)}\right)$$

$$= \sum_{k=2}^n \left[\ln(k^2) - \ln[(k-1)(k+1)] \right]$$

$$= \sum_{k=2}^n \left(2\ln k - \ln(k-1) - \ln(k+1) \right)$$

$$\therefore \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < \left[(2\ln 2 - \ln 1 - \ln 3) + (2\ln 3 - \ln 2 - \ln 4) \right. \\ \left. + (2\ln 4 - \ln 3 - \ln 5) + \dots + 2\ln(n) - \ln(n-1) - \ln(n+1) \right]$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 1 + \ln 2 + \ln\left(\frac{n}{n+1}\right)$$

$$\therefore \sum_{k=1}^n \frac{1}{k^2} < 1 + \ln 2 + \ln\left(\frac{n}{n+1}\right)$$

$$\text{As } n \rightarrow \infty, \frac{n}{n+1} \rightarrow 1 \therefore \ln\left(\frac{n}{n+1}\right) \rightarrow 0$$

$$\therefore \sum_{k=1}^{\infty} \frac{1}{k^2} < 1 + \ln 2.$$

End of paper